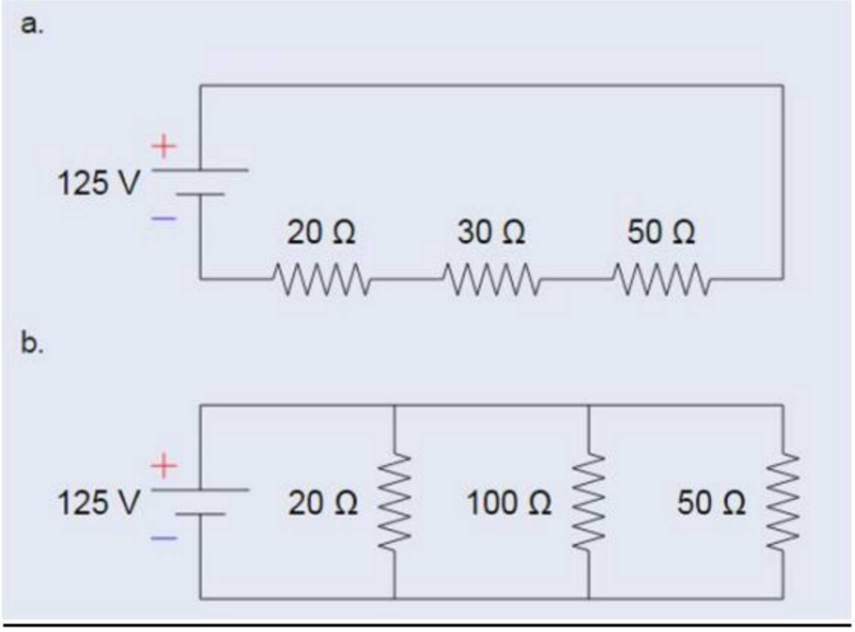
IOT assignment 2



For each of these circuits:

1. Calculate the total resistance in ohms,
   a. 20 + 30 + 50 = 100ohms
   b. 1/20 + 1/100 + 1/50 = 12.5ohms
2. Calculate the total current from the power supply,
   V = IR, SO I = V/R
   a. 125/100 = 1.25A
   b. 125/12.5= 10A
3. Calculate the current through each resistor
   a. 1.25A since current in a series circuit is the same everywhere
   b. Voltage is the same, so I = V/R
      i. 1- 125/20 = 6.25A
      ii. 2- 125/100 = 1.25A
      iii. 3- 125/50 = 2.5A
4. Calculate the Voltage drop through each resistor
   a. V=IR
      i. 1- 1.25 × 20 = 25 V
      ii. 2 - 1.25 × 30 = 37.5 V
      iii. 3 - 1.25 × 50 = 62.5 V
   b. Voltage is the same all across so 125V
5. Calculate the Power dissipated through each resistor (use Watt's Law: Power = Voltage * Current)
   a. P= VI
      i. 1 - 25 × 1.25 = 31.25 W
      ii. 2 - 37.5 × 1.25 = 46.875 W
      iii. 3 - 62.5 × 1.25 = 78.125 W

b.

i. 1 - 125 × 6.25 = 781.25 W
ii. 2 - 125 × 1.25 = 156.25 W
iii. 3 - 125 × 2.5 = 312.5 W